

Linework in Spider-Man Across the Spider-Verse

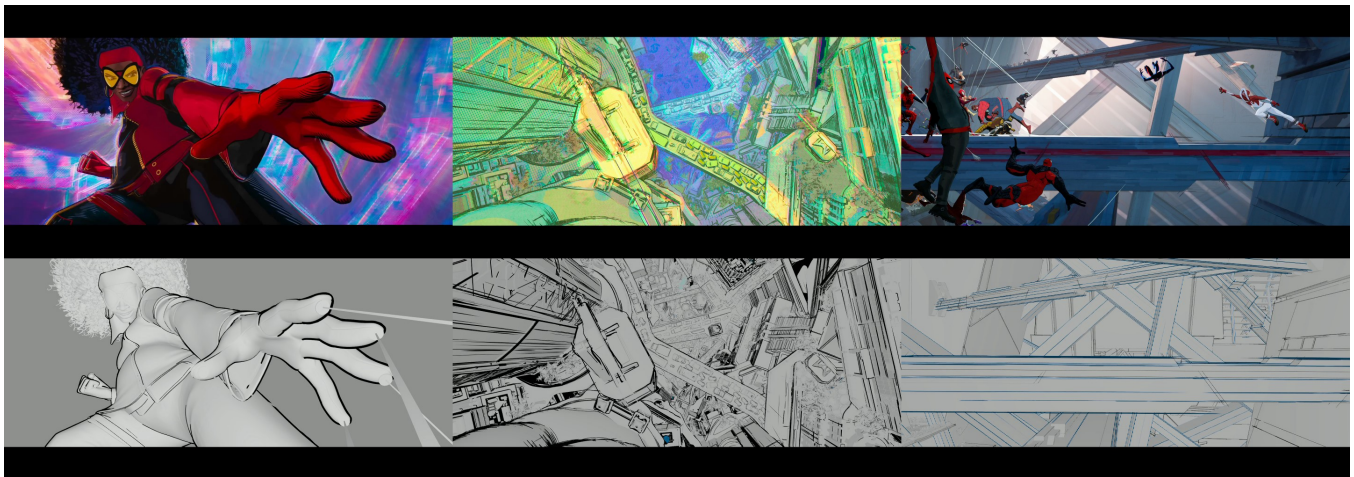
An artistic driven approach to linework generation

Pav Grochola
pgrochola@imageworks.com
Sony Pictures Imageworks
USA

Young Joon Lee
yjlee@imageworks.com
Sony Pictures Imageworks
Canada

Filippo Maccari
fmaccari@imageworks.com
Sony Pictures Imageworks
Canada

Edmond Boulet-Gilly
ebouletgilly@imageworks.com
Sony Pictures Imageworks
Canada



©2023 CTMG, Inc. All Rights Reserved

Figure 1: Examples of different types and style of line-work in *Spider-Man: Across The Spider-Verse*.

ABSTRACT

In *Spider-Man: Across The Spider-Verse* we present an innovative approach to generating artistic linework for animated 3D films. This technique allows for extensive artistic freedom, explicit animation and redrawing of the line-work, while also seamlessly adhering to the underlying animation.

CCS CONCEPTS

• **Computing methodologies** → *Procedural animation; Physical simulation; Non-photorealistic rendering.*

KEYWORDS

Production, Pipeline, Comic Book, Ink Lines

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595456>

ACM Reference Format:

Pav Grochola, Filippo Maccari, Young Joon Lee, and Edmond Boulet-Gilly. 2023. Linework in Spider-Man Across the Spider-Verse: An artistic driven approach to linework generation. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595456>

1 INTRODUCTION

In *Spider-Man: Across The Spider-Verse* the significance of incorporating linework was essential because each world in the project is rooted in five unique comic book styles. The demand for line-work grew substantially, as we aimed to apply lines to both characters and environments in their entirety. We took great care in considering how to approach this challenge.

2 FEATURES

We identified several key features that a versatile linework solution must possess for our production:

- A high degree of artistic control to fine-tune the appearance of the lines, with each world and character having its unique linework style.

- The linework should look as if it were hand-drawn, often extending beyond the surface for characters with a messy appearance.
- The linework must redraw itself convincingly, updating on-the-fly rather than remaining static on the geometry. Furthermore, it was also crucial that the linework animation was synchronized with the character animation's stepped timing.
- We needed an efficient way to save line-work styles and then batch process the generation of linework efficiently with minimal effort.
- The system should combine hand-made linework with procedurally generated ones.

3 DESIGN AND IMPLEMENTATION

3.1 Kismet

We developed a tool called "Kismet" in Houdini to address these features. Although capable of generating a broad range of linework styles (including hatching), its primary use was as a versatile outliner. We deconstructed the problem into several steps:

- (1) We parsed the scene data and analyzed where lines could appear based on the camera angle relative to the 3D mesh and generated a base curve for all potential line locations.
- (2) Based on user input, we then produced secondary lines on top of the base curve. These lines followed the animation of the base curves and ultimately got rendered.
- (3) The secondary lines were "stylized" meaning that we assigned various properties such as varying tapering and offsetting to achieve a complex, natural look.
- (4) The curves were then output to Katana for rendering

3.2 The hand-made X-Factor

While Kismet generated convincing linework, we discovered that combining it with hand-drawn lines yielded the best results. On certain occasions, artists utilized Blender to create drawings that were exported to Houdini, while in other situations, 2D FX artists exported vector data from their DCC applications. These sets of drawings were then interpolated and merged with the curves generated by Kismet.

4 COMPLEXITIES OF ANIMATION

Although replicating the intricate and organic appearance of hand-drawn ink lines was a significant challenge, our toolset provides comprehensive options for defining how the linework behaves on the animated models.

As lines are "drawn on" the curves gradually reveal their opacity, much like an invisible pencil sketches a line on paper. Likewise, when lines are "erased off" the curves lose their opacity, as if being gradually wiped away. The rate at which this process occurs, encompassing both the appearance and removal of lines, is known as the "redraw rate."

Determining the optimal "redraw rate" proved tricky. When a character remains relatively stationary or still, a high redraw rate would prove distracting. Conversely, if the redraw rate is set too low for a moving character lines would get stuck on the model.

This raises issues for outlines as they would appear disconnected from the silhouette.

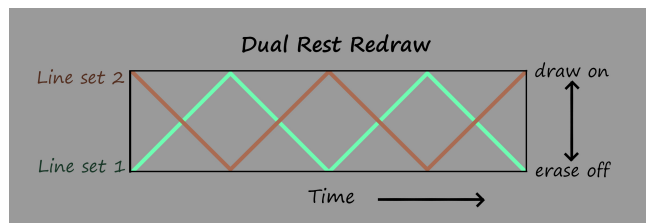
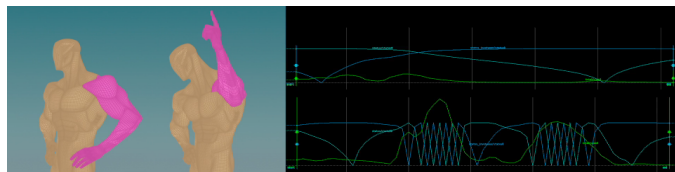


Figure 2: Dual rest redraw

To overcome this problem we applied a "dual rest" approach, whereby two sets of lines work in tandem, with one set "draw on" while the other set is "erased off" and vice versa. (see Figure 2)

We found that having two sets of lines with a correct re-draw rate resulted in a natural looking linework update.

We took this concept to the next level by creating a system that automatically detects the degree of change in a character's performance, enabling us to establish the appropriate redraw rates across different parts of the character.



©2023 CTMG, Inc. All Rights Reserved

Figure 3: Left: Character partitioned in multiple parts based on screen space angular velocity. Right: We analyze the activity of each partition.

5 PRODUCTION READY PIPELINE

Populating the show with ink lines proved to be a challenging task, as we had to generate linework for more than 1000 shots. To manage this workload, we established a separate department of about 30 artists whose sole responsibility was to create ink lines for the show. This not only alleviated the burden on the FX department but also allowed the artists to specialize in and become proficient in this role. We also developed an extensive templates-based workflow to easily deploy ink lines generation through command line.

ACKNOWLEDGMENTS

Special thanks to our Production, Systems, Dev and PSR team members, past and present, including Steve Vargas, Sosh Mirsepassi.

REFERENCES

- Harold Speed. 1972. *The Practice and Science of Drawing*.
Walt Disney Animation Studios. 2012. Paperman.